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DRAG AND DROP

Playwright + TypeScript

Drag and Drop is a common UI interaction where you click on an element, hold it, drag it to another location, and release it. Playwright makes this super easy with the `dragTo` method!

IMPORTANT!

`dragTo` method handles the entire drag and drop operation in one line! It clicks the source, drags it, and drops it on the target automatically.

Understanding Drag and Drop

How it works:

Before Drag:

SOURCE

TARGET

Drag and Drop

After Drop:

(empty)

SOURCE

The dragTo Method

Playwright provides the dragTo method to perform drag and drop operations!

Syntax

```
await sourceLocator.dragTo(targetLocator);
```

What dragTo does:

Clicks on the source element

Holds the mouse button down

Moves the element to target location

Releases the mouse button

All in ONE method call!

Real Example - jQuery UI Droppable

Let's drag a draggable element and drop it on a droppable area!

Complete Code

```
import { Browser, chromium, expect, Page } from "playwright";

let page: Page;

(async () => {
  const browser: Browser = await chromium.launch({
    headless: false,
    channel: "chrome",
  });

  page = await browser.newPage();
  await page.goto('https://jqueryui.com/resources/demos/droppable/default.html');

  // Step 1: Locate source element
  let source = page.locator('#draggable');

  // Step 2: Locate target element
  let target = page.locator('#droppable');

  // Step 3: Drag source to target
  await source.dragTo(target);

  // Step 4: Verify drop was successful
```

```
await expect(target).toContainText('Dropped!');  
})();
```

Breaking it down:

Step 1: Find the element to drag (source)

Step 2: Find where to drop it (target)

Step 3: Use dragTo method

Step 4: Verify with assertion

Another Example - Drag and Drop Columns

Drag and Drop Columns

```
await page.goto('https://the-internet.herokuapp.com/drag_and_drop');  
  
let source = page.locator('#column-a');  
let target = page.locator('#column-b');  
  
// Drag column A to column B  
await source.dragTo(target);  
  
// Verify column A is now in position B  
await expect(target).toContainText('A');
```

Step-by-Step Process

Complete Drag and Drop Flow:

1. Navigate to the page
2. Locate the source element (what to drag)
3. Locate the target element (where to drop)
4. Call `dragTo` method
5. Verify with expect assertion

Key Points About `dragTo`

`dragTo` is a single method that does everything

No need for separate `mouseDown`, `mouseMove`, `mouseUp`

Works with HTML5 drag and drop

Automatically waits for elements to be ready

Simple and clean syntax!

REMEMBER!

Use dragTo for simple drag and drop operations

Always locate both source and target elements first

Add assertions to verify the drop was successful

Playwright handles all the complex mouse movements automatically!

COMMON MISTAKES

Not waiting for elements to be visible before dragging

Using wrong locators for source or target

Forgetting to verify the drop was successful

Not handling elements inside iframes properly

PRO TIPS!

Always add assertions after drag and drop

Use unique locators (ID, test-id) for source and target

Playwright auto-waits, no need for manual waits

Test on different browsers - drag and drop behavior can vary

Real-World Use Cases

1. Kanban boards - Moving tasks between columns
2. File uploads - Dragging files to upload area
3. Reordering lists - Changing item positions
4. Dashboard widgets - Rearranging components
5. Image galleries - Organizing photos

KEY TAKEAWAYS

Use `dragTo` for drag and drop operations

Syntax: `await source.dragTo(target)`

Handles all mouse movements automatically

Always verify with assertions

Simple, clean, and powerful!

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